Samson Wu

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EDUCATION

Northeastern University — Boston, MA
Candidate for Master of Science in Robotics (Khoury College of Computer Sciences) — Sep 2026 – Present
Related courses: Reinforcement Learning, Robotic Sensing and Navigation — Expected Graduation: Dec 2028
Focus: Mobile/Loco-manipulation and dexterous manipulation of robots

University of Toronto — Toronto, ON
BASc in Engineering Science, Major in Machine Intelligence — Sep 2021 – May 2026
Honours: Dean's List (7 semesters) — GPA (Last 2 years): 3.78
Related courses: Machine Learning, Control and Robotics, Computer vision

TECHNICAL KNOWLEDGE

Languages: Python, Java, SQL, C
Software: Mujoco (Robocasa), Pytorch, Tensorflow, ROS2, GCP, AWS, PyCharm, DBVisualizer, SSMS, Apache Airflow, Jira, Postman
Other Skills: JAX, Fusion360, MATLAB, PySpark, Springboot, React, T-SQL, Devops and CI/CD

EXPERIENCE

AI Developer @ Geotab, Toronto, ON — May 2026 – Aug 2026

- Optimized workflow, improved UX, and increased determinism for internal AI Agent SDLC workflow (Claude Code plugin), overall increased development speed by 5-10 times compared to humans.
- Engaged in rapid development and support for a multi-cli internal AI harness (runtime) tool used by the entire organization daily for enhanced productivity. (~2000 monthly active users, 60% of organization)
- Designed and implemented a pipeline for automatic skill demotion using GCP and Gitlab infrastructure, eliminating hours of weekly manual labour.

Data Platform Developer @ Geotab, Toronto, ON — May 2024 – Apr 2025

- Engineered a distributed Ray pipeline to parallelize heavy video frame ingestion, validating scalable asynchronous workload processing for high-throughput video pipelines.
- Developed, fixed bugs, vulnerabilities, and wrote unit tests for a microservices-based Java Spring Boot video processing cloud application in GCP to add features and increase code coverage.
- Designed and implemented automated monitoring cronjobs deployed in the cloud using SQL and Python to measure performance of the application and provide alerting in production.

PROJECTS

Autonomous robot for unloading and loading a dishwasher, Thesis — Sep 2025 – May 2026

- Conducting a comprehensive survey on mobile manipulation frameworks, spatial representations, and robot learning paradigms to establish baseline architecture.
- Evaluated, and trained multiple sample-efficient Point-cloud models in Pytorch to predict rack occupancy for dishwasher loading and unloading, achieving 80%+ accuracy on custom synthetic dataset of 1200+ RGBD.
- Applied research to integrate all systems (localization, rack modelling, CNN model, Vision Foundation models, grasp candidates) in an autonomous pipeline to achieve 70% load success rate in simulation.

INTERESTS

- Rubik's Cubing: Former top 132 in Canada for 3x3 one-handed single time — Mar 2020
- Volleyball: 1st in U15 Div. 2 Ontario Championships — Apr 2018
- Art: 2nd in painting competition (300+ participants)